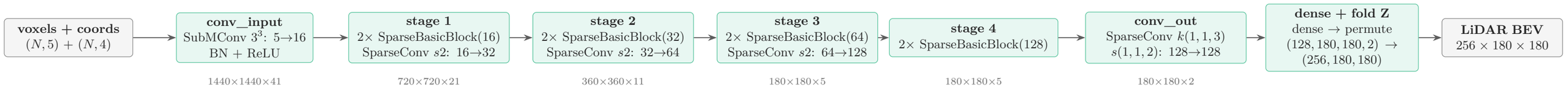
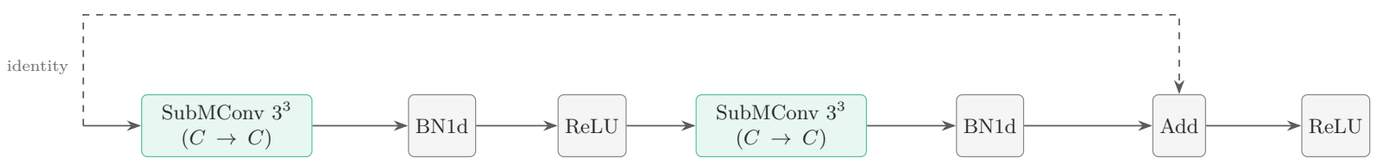
SparseEncoder: 3D sparse convolution backbone (SparseConv3d / SubMConv3d)



SparseBasicBlock (residual, used 2× per stage)



Sparse tensors carry features (N, C) + indices $(N, 4)$; every conv keeps the sparse index map. Strided $s2$ convs sit at the end of stages 1–3; stage 4 is submanifold-only.

Coordinates are (batch, x, y, z) with spatial shape 1440 × 1440 × 41 (det: xy 0.075 m, z 0.2 m) or 1024 × 1024 × 41 (seg: xy 0.1 m, z 0.2 m). Stage strides halve x/y/z (41→21→11→5), conv_out halves z again (5→2); Z is folded into channels, so the LiDAR BEV feature is 128×2 = 256 channels.